

Sadhan Kumar Ghosh *Editor*

Urban Mining and Sustainable Waste Management

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Contents

Contribution of Optimal Equipment for ELV Recycling to the Sustainable Development of Serbia and the Region	1
M. Pavlović, D. Tadić, A. Aleksić, N. Komatina and B. H. Hosam	
A Study on Biomedical Waste Management in Chittoor District	9
N. Hanumaiah, P. Lakshmi Narayana Prasad and J. Karthikeyan	
Knowledge and Practice of Biomedical Waste Management and Awareness of 3 ‘R’s Concept Among Staff Nurses in the Hospital—A Cross-sectional Study	15
Bonam Soujanya Kumar, Md. Sameer, S. B. Vishnu murthy and G. S. C. N. V. Prasad	
Utilization of Construction and Demolition (C&D) Waste and Industrial Inorganic Wastes in Cement Manufacturing	27
Parmanand Ojha, Pinky Pandey, Kåre Helge Karstensen and Palash Kumar Saha	
A Review of Studies on Environmental Performance Analysis of Construction and Demolition Waste Management using Life Cycle Assessment	39
Kishore C. Kumar, V. G. Ram and Satyanarayana N. Kalidindi	
Extraction of Selected Metals from High-Grade Waste Printed Circuit Board Using Diethylene Triamine Penta-acetic Acid	49
Auchitya Verma, Amber Trivedi and Subrata Hait	
A Comparative Study on the Cost–Benefit Analysis on Metal Recovery of WPCB Using Pyrometallurgy with Two Different Thermal Furnaces	59
R. Balaji and J. Senophiyah-Mary	

Recycling of Polymers from WEEE: Issues, Challenges and Opportunities	69
Biswajit Debnath, Ranjana Chowdhury and Sadhan Kumar Ghosh	
Slate Mine Wastewater, the Best Substitute for Cementation	81
S. Altaf Hussain, S. M. Subhani and S. V. Satyanarayana	
Waste Coal Utilization in India: A Review	91
Krishna Kant Dwivedi, Prabhansu, M. K. Karmakar, A. K. Pramanick and P. K. Chatterjee	
Temporal Changes of Solid Waste at Limestone Quarries in and Around Yerraguntla, YSR District, A.P., using Google Earth Images	99
Y. Sudarshan Reddy, B. Suvarna, M. Prasad, V. Sunitha and M. Ramakrishna Reddy	
Waste Is not a Waste—It's Time to Realize	111
M. Rao Divi	
Effective Treatment for COD Removal of Landfill Leachate by Electro-coagulation	129
P. T. Dhorabe, A. R. Tenpe, V. S. Vairagade, Y. D. Chintanwar, B. R. Gautam and V. R. Agrawal	
Estimation of Greenhouse Gas Emissions from Matuail Landfill Site	149
Md. Maruful Hoque and M. Tauhid Ur Rahman	
Stabilization of Contaminated Soil in a Landfill Site with Ground Granulated Blast Furnace Slag	161
Ramiz Raja and Supriya Pal	
Air Quality Survey of Some Major Dumpsites in Lagos State, Nigeria	169
Omowonuola Olubukola Sonibare, Adeniyi Saheed Aremu, Rafiu Olasunkanmi Yusuf and Jamiu Adetayo Adeniran	
Remediation of bis(2-Ethylhexyl) Phthalate and Phenol, 4,4'-(1-Methylethylidene)bis—in Landfill Leachate Using Biopolymer	185
P. Agamuthu, A. Aziz, A. Hassan and S. H. Fauziah	
Biodegradation of Plastic Waste Using Marine Micro-Organisms	195
Rwiddhi Sarkhel, Shubhalakshmi Sengupta, Papita Das and Avijit Bhowal	
Current Scenario of Plastic Waste Management in India: Way Forward in Turning Vision to Reality	203
Tadinada Sri Sasi Jyothsna and Bandari Chakradhar	

Modelling of Post-consumer Plastic Flow in Municipal Solid Waste Stream: A Case Study in Few Major Local Authorities of Sri Lanka	219
Yushani Alahakoon and Anurudda Karunarathna	
Youth Engagement for E-waste Management by Urban Local Bodies in India	231
Sai Dinesh Shetty	
Improvement in Engineering Behaviour of Expansive Soil Reinforced with Randomly Distributed Waste Plastic Strips	239
Rachita Panda and Sudhira Rath	
Willingness of Students and Academicians to Participate in E-waste Management Programmes—A Case Study of Bangalore	249
Khushbu K. Birawat, Biswajit Debnath, Shushmitha. L. Gowda and Sadhan Kumar Ghosh	
Purification Technologies of Bioreactor Landfill Gas and Its Sustainable Usage: Current Status and Perspectives	263
Anaya Ghosh, Jyoti Prakas Sarkar and Bimal Das	
NO_x Reduction from Diesel Engine Consuming Diesel-Waste Plastic Oil Blend as Substitute Fuel with Antioxidant and SCR	273
B. Sachuthananthan, R. L. Krupakaran, A. Sumanth and T. Manikandan	
A Study of Solid Waste Management in Indore City: With Special Reference to Biomedical Waste	287
Mohini Jadon	
Hydrothermal Carbonization—A Sustainable Approach to Deal with the Challenges in Sewage Sludge Management	293
Vicky Shettigondahalli Ekanthalu, Gert Morscheck, Satyanarayana Narra and Michael Nelles	
Sustainable Bio Medical Waste Management—Case Study in India	303
Vandana, V. Venkateswara Rao and Sadhan Kumar Ghosh	
A Study on Plastic Waste Management by Stakeholders Using Reverse Logistics	319
R. Madhura	
A Study on Pyrolysis of Plastic Wastes for Product Recovery and Analysis	329
Ankita Mukherjee, Biswajit Ruj, Parthapratim Gupta and A. K. Sadhukhan	

Temporal Changes of Solid Waste at Limestone Quarries in and Around Yerraguntla, YSR District, A.P., using Google Earth Images



Y. Sudarshan Reddy, B. Suvarna, M. Prasad, V. Sunitha
and M. Ramakrishna Reddy

Abstract In limestone mining and cement industry, solid and liquid wastes are generated in every day and every stage of the operations and are required to mitigate properly. Different types of waste generated from both the industries are cement and limestone quarry. Due to environmental impact and public health and safety, proper efforts must be made to decrease waste generation and hence efficient disposal practices have to be followed. Hence, this study aims at a better understanding of spatial and temporal changes of unplanned dumping sites from 2006 to 2018. Google Earth mapping is one of the most advanced methods for identification of rock-solid waste clearly for collecting on satellite image data. This paper is centered on application of Google Images in assessing the temporal changes of solid waste at limestone quarries in and around Yerraguntla. Time series multi-date Google Earth imageries of 2006–2018 are used to demarcate the evolutionary changes in limestone waste disposal management and to understand the spatial and temporal changes that happened due to the changes by expanding the rock waste dump around Yerraguntla village, YSR district, A.P. Results revealed that the solid waste management in the study area is very poor which need to be properly monitored so as to mitigate the present and future environmental threats.

Keywords Temporal changes · Solid waste · Limestone quarries · Google Earth · Yerraguntla

1 Introduction

Waste is defined as the discarded and discharged material generated during every stage of life causing adverse health and environmental impact (Bringi 2007). Environmental contamination and waste management are the major concerns to earth scientist and form other related fields of science all over the world both in developing

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and in developed countries (Soupios et al. 2004). The major factors which contribute solid waste generation are growing population, rising economic growth, and rise in community living standards. Limestone solid wastes represent huge masses of bulk materials that are often deposited without any consideration, causing a lot of difficulties and inviting illegal deposit of other kinds of waste and mine garbage (Kartam et al. 2004). As a result, the building and construction industry play an important role in waste generation. At present, waste management became an unavoidable challenge in regional level, national level, and global level. Mining is one of the major incoming sources for the developing countries like India, and at the same time it is the major source of huge amounts of solid waste generation in terms of overburden, milling, processing, and tail end wastes which need to be monitored and maintain proper mine plan and environmental policy particularly in opencast mining areas. The unplanned mining became a serious impact on the environment by unplanning mining activity causing reduction of forest cover, soil erosion, pollution air, water, land and reduction of biodiversity in earth surface. The waste rock dumps could alter landscape around mining areas (Mondal et al. 2013; Sarma and Kushwaha 2005). Mining activity also results in environmental impact with respect to human settlements, degrading air, water quality, forest land, etc. (Dumitru et al. 2016). In India, limestone mines are operated opencast mining method (Indian Minerals Yearbook 2017). When compared to underground mining, opencast mining activity causes adverse impacts; hence, proper environmental management plan has to be adopted. In order to ensure long-term stability of storage and dispersed facilities and to minimize air, water, and land contamination, proper management technique is needed. If the Mine water is good enough quality it can be used for public supply but when the management failure of control solid waste harmful impacts, such as depletion of water resources by dewatering, or the pollution of surface water courses by poor quality mine waters and mine waste leachates, demand careful inspection (Younger and Wolkersdorfer 2004). The inappropriate or unsafe management of solid rock wastes at quarry working continues to create opposition from local communities and the general public and has contributed to the negative impact on public perception of the mining industry. It can be better monitored by the remote sensing satellite data than the underground mining. To identify suitable sites for mining and also take a remedial measure guidelines approved by the Ministry of Environment Forests and Climate Change (MoEF and CC 2008) and Indian Bureau of Mines (IBM) can reduce the major possible impacts. In a present scenario where the mineral extraction was done on a large scale, it became a highly subject of concern regarding the limestone quarry waste being generated by this limestone mineral extraction, which has a huge impact on the environment; therefore, a regular study should be taken in accordance with the application of latest technology in order to sustain this earth (Das and Choudhury 2013). The Google Earth (GE) shows the area at fine spatial resolution helpful clearly identified waste dumps surrounding the limestone quarries. Using Google Earth images identifies dumps in limestone quarry locations to better understand the effect of mines on vegetation, air, and groundwater disturbances (Jade and Sunitha 2015). If detailed change detection is needed for a study, post-classification will be

also needed (Zhang et al. 2017). This research used Google Earth data detection solid waste changes near Yerraguntla town (Wibowo et al. 2016).

2 Study Area

Yerraguntla is an industrial village in YSR district in Andhra Pradesh. Geographically, Yerraguntla is located at 14.6333°N 78.5333°E with average elevation of 152 meters (501 ft). It is 37 km away from district head quarter Kadapa and well connected with rail road. The study area is well known for various cement industries like Zuari cement Ltd, Bharathi cement Ltd, and has good number of stone polishes industries, besides Rayalaseema Thermal Power Plant was established. Niduzivi, Koduru, Valasapalli areas of Yerraguntla Mandal are known for Napa slabs/Kadapa slabs are black colored limestone belonging to Koilakuntla Limestones (Sunitha et al. 2014). YSR district occupies lion share of limestone reserves in Andhra Pradesh, which is the major state among the limestone-producing states with 31, 96,000 ton production as per Indian Mineral Yearbook (2017) (Figs. 1 and 2).

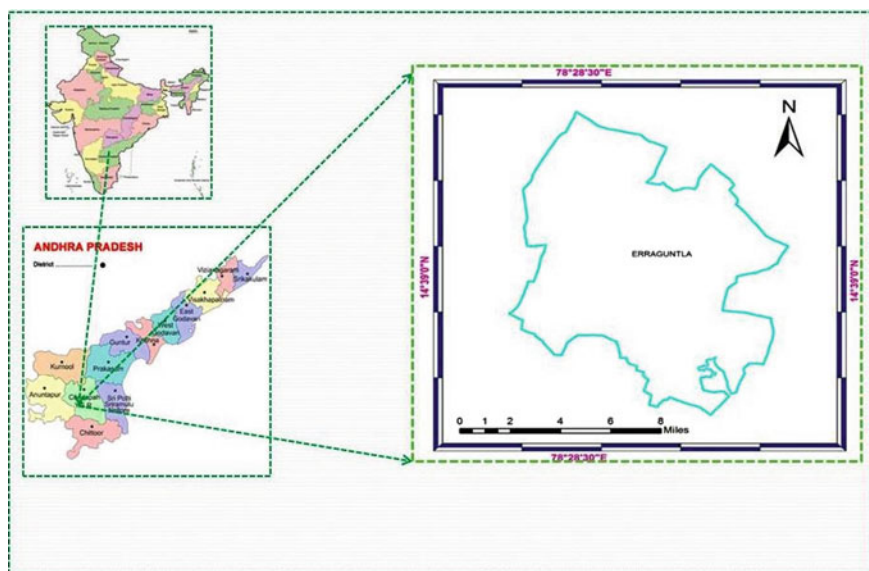


Fig. 1 Location map of the study area

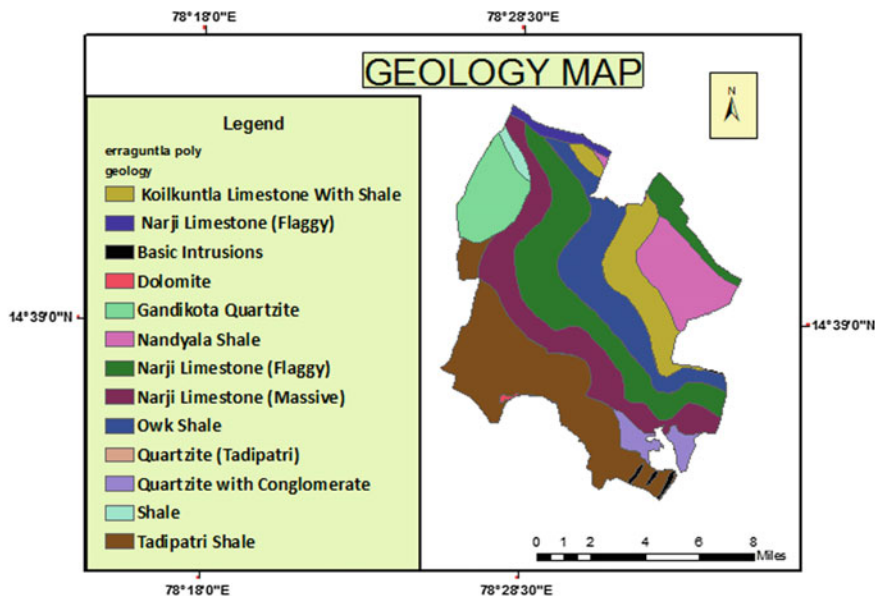


Fig. 2 Geology map of the study area

3 Methodology

A systematic integrated methodology is adopted for the present study to assess the temporal changes in waste disposal. The intensive study of the relevant literature, periodicals, etc., has been carried out; based on the review of the literature, it is clear that remote sensing is the only effective tool to monitor the changes for multi-date scenarios. Thus, Google Earth satellite images were chosen as the data source for the study area and the time series changes have been mapped for the years 2006 and 2018. Field work has been carried out in the study area like waste disposal sites, mines, tailings, cement industries, stone cutting and polishing industries to know to type of waste generation, handling, recycle and disposal. Data generated from the Google Earth was compared and verified ground truth data by GPS survey, topographic maps, field photographs, and questioner from local sources and mine and industrial management (Figs. 3 and 4).

4 Results and Discussion

Limestone mining waste disposal locations were identified on Google Earth imagery, based on the image elements like tone, texture, shape, size, pattern, and association. The active dumps are clearly visible gray tone with medium to coarse texture due

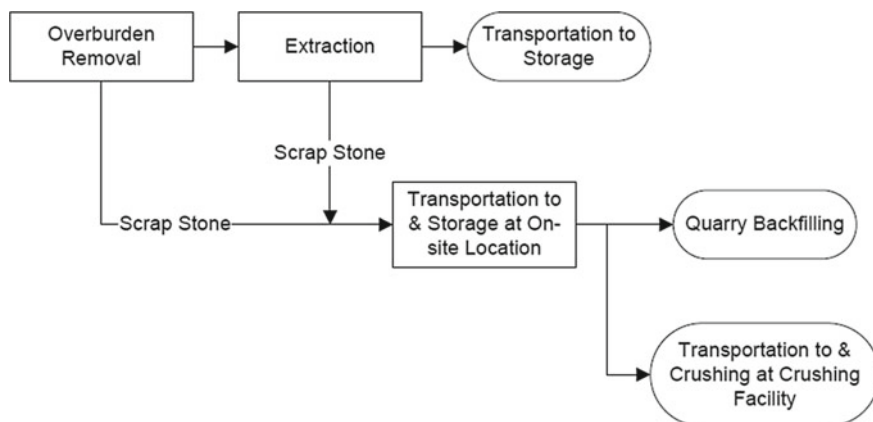


Fig. 3 Flowchart of the limestone quarry processing

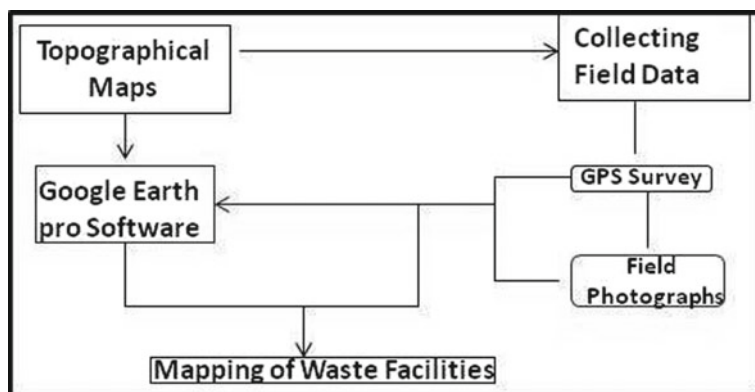


Fig. 4 Flowchart of the methodology

to the presence of debris of waste and overburden and irregular in shape and size, whereas the mine pits are with perfect shape and mainly in dark blue tone due to the presence of water in the pits as shown in Figs. 5 and 6.

In mining and cement industry, wastes are generated in every stage of the operations; in the present study, we have identified two major locations of solid waste disposal: One is near the mine, and the second one is besides the road. Spatiotemporal changes in 2006–2018 are clearly depicted by drawing the red-colored polygon for both the locations (Figs. 5a–f and 6a–f).

By observing the Google Earth temporal images as shown in Figs. 5 and 6 and the field observations as shown in Figs. 7, 8, 9 and 10, it is clear that overburden management and solid waste of these quarries are extremely poor; most of the mining companies are poor in practices on overburden management and fugitive dust from waste pile. They do not bother to take preventive measures against the limestone waste



Fig. 5 Temporal and spatial formation and changes during 2006–2018 in location of solid waste beside railway track surrounding Yerraguntla village

management. The best method is to backfill the excavated land, and this is hardly implemental as companies keep opening from time to time different faces of mine open began without exhausting previous one. The opencast mines are characterized by enormous open pits surrounded by huge dump yards surrounded by the croplands. Dust and slurry (Figs. 11 and 12) from these sites are being transported by the air and rainwater to nearby cropland ponds which is major threat to soil porosity, productivity, permeability, and crop yield. During dry summer, these dumps become a key source of air pollution for the surrounding areas.

5 Limestone Waste Properties

Diverse stone wastes like granite, quartzite, limestone, marble, each one has different properties. Limestone waste may have different physical properties depending on the extraction of mineral and method of generation of waste. The properties of limestone waste used as an aggregate in concrete are presented in Table 1 (Lakhani et al. 2018).

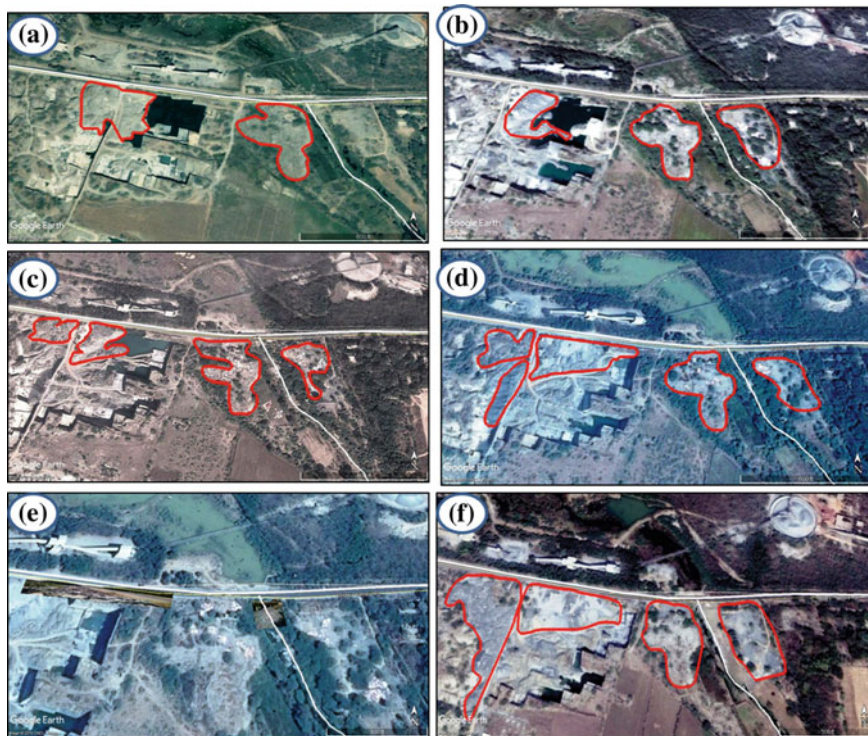


Fig. 6 Temporal and spatial formation and changes of solid waste during 2006–2018 beside high-way road near India cement factory surrounding Yerraguntla village

Fig. 7 Limestone dump near India cement factory, Yerraguntla village



Fig. 8 Limestone surrounding Yerraguntla village



Fig. 9 Overburden of Niduzuvvi limestone quarry surrounding the Yerraguntla village



6 Conclusion

The conclusion of this research on the spatial–temporal change detection using the Google Earth data is very important, low cost, and very useful as spatial–temporal detection for solid waste investigation, monitoring, and mapping in mining industries. The present study intended to assess the spatiotemporal changes in the disposal of solid waste derived from the different types and the quantities of mining waste, in and around Yerraguntla village area in Kadapa district by using the Google Earth imagery. Based on the satellite data analysis and ground truth verification, it is concluded that overburden and the tailings from the limestone are extremely poor and there is no strict control on the solid waste disposal and management plan which resulted in the dumping of waste in the nearby streams, roadside, and the banks of the highway.

Fig. 10 Solid waste in a stream near India cement factory, Yerraguntla village



Fig. 11 Water inundation in limestone quarry surrounding Yerraguntla village



Fig. 12 Development of fine dust during transport surrounding Yerraguntla village



Table 1 Physical and mechanical properties of solid and powder limestone waste

S. no.	Property					
	Type of stone waste	Specific gravity	Bulk density (kg/m ³)	Water absorption (%)	Maximum size (m/mm)	Color
1.	Limestone	2.6–2.65	1568–1743	2–4	2 m (solid) 0.2 mm (powder)	Gray

Finer material from Limestone quarries had been carried out by rain water into near by water bodies or lands and pollutes both media. It also observed that dust emissions from the mine and dumps become a key source of air pollution for the surrounding areas.

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